

COMP 550 - Natural Language Processing - Final Project Proposal
Cross-Lingual Sentiment Disparity toward Geopolitical Adversaries

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Introduction

Multilingual language models are widely used for sentiment analysis, but recent work suggests they may carry language-dependent biases. Because these models are trained on corpora drawn from many regions, their predictions may vary depending on the language a prompt is written in. We want to test whether multilingual models show a "Geopolitical Language Gap": do countries labeled as geopolitical adversaries receive more negative sentiment predictions in English than in their native languages? We also expect this gap to be larger for adversary nations than for neutral or allied ones.

Related Work

Barbieri et al. (2022) introduced the XLM-T model family, trained on large-scale Twitter data for sentiment analysis, and provide the model we use. Faisal and Anastasopoulos (2022) showed that uneven representation of regions in training data leads to disparities in model behavior. Li et al. (2024) showed that multilingual models produce language-dependent inconsistencies on geopolitical questions. Our project zeros in on sentiment predictions toward geopolitical entities, using controlled multilingual prompts.

Method

We use twitter-xlm-roberta-base-sentiment (Barbieri et al., 2022), a transformer classifier trained on ~198M tweets and fine-tuned on the UMSAB sentiment dataset. We extract the raw softmax probabilities for the "Negative" class and perform a two-way ANOVA to test the interaction effect between language and country category on sentiment output.

Our inputs are synthetic "mirror templates," neutral sentences referencing a country (e.g., "I saw a news report about [Country]"). We will construct 30 templates and translate them into English, Chinese, Russian, and Arabic, selected because they correspond to major geopolitical actors and can be verified by team members. We analyze 30 countries across three categories (10 adversaries, 10 allies, 10 neutral), operationalized using the U.S. National Security Strategy (2022), NATO membership, and non-aligned status respectively.

For each template, language, and country, we compute the sentiment score and average across templates per country–language pair. Our primary measure is the difference between a country's English score and its native-language score, compared across geopolitical categories. Results will be summarized with category-level averages, heatmaps of sentiment scores by country and language, and ANOVA interaction plots.

Contributions and Limitations

The main goal is to empirically test whether cross-lingual sentiment disparities exist in a widely used multilingual model. One limitation is the use of synthetic templates, which may not reflect real-world discourse; if feasible, we will supplement with real multilingual tweets. Geopolitical categorizations based on Western frameworks may also introduce bias, which we will note when interpreting results.

Statement of Contributions

Nabil: template design, Arabic translation verification, report writing

Maximillian: inference pipeline implementation, data extraction

Sebastien: statistical analysis, visualizations, Chinese/Russian verification

References

Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa Anke, L. (2022). TweetEval: Unified benchmark and comparative evaluation for tweet classification.

Faisal, F., & Anastasopoulos, A. (2022). Geographic bias in NLP systems.

Li, X., et al. (2024). BorderLines: Language-dependent inconsistencies in multilingual language models regarding geopolitical facts.